Does bleeding affect patient-reported outcome measures in patients with myelodysplasia or hematologic malignancies: a systematic review

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BACKGROUND: Relatively minor bleeding (e.g., bruising and/or petechiae) may cause patient distress. This systematic review's objective was to assess whether bleeding affects health-related quality of life (HRQoL) or illness perceptions or representations (IPs) in patients with hematologic malignancies or myelodysplasia (MDS).

STUDY DESIGN AND METHODS: We searched, in full, 12 electronic databases (including CENTRAL, MEDLINE, and EMBASE) up to January 7, 2013, for eligible randomized controlled trials (RCTs), prospective cohort studies, and cross-sectional studies.

RESULTS: A total of 6247 studies were initially identified; 5945 studies were excluded on the basis of the abstract. A total of 302 full-text articles were evaluated independently by two reviewers; of these, six studies within seven citations were eligible for inclusion. Two studies are still in progress, four studies within five citations were included in this review (one RCT, one prospective observational study, one interview study, and one Web-based survey). None of the included studies were designed to assess the impact bleeding had on HRQoL or IPs. The Web-based survey and observational study used two new patient-reported outcome scales which specifically assessed patient distress or concern due to bleeding. The majority of patients within these two studies either did not experience bleeding or did not have severe thrombocytopenia.

CONCLUSION: There is insufficient evidence to demonstrate whether bleeding is a significant clinical problem that affects patients' HRQoL or IPs in either patients with MDS or patients with hematologic malignancies. Rigorously designed studies to assess the scale of this problem in both of these groups of patients are required.

Clinicians may not realize that relatively minor bleeding could cause patient distress. Petechiae are seen by doctors as a relatively trivial manifestation of thrombocytopenia but patients could see them as an outward sign of their illness and therefore distressing.

Any type of supportive care should explicitly assess quality of life.[1] The majority of platelet (PLT) transfusions (69%) given to patients with hematologic malignancies or myelodysplasia (MDS) are to prevent bleeding (i.e.,

ABBREVIATIONS: FACT = Functional Assessment Cancer Therapy; HRQoL = health-related quality of life; IP(s) = illness perceptions or representation(s); ITP = immune thrombocytopenia; ITPA = interactive tailored patient assessment; MDASI = MD Anderson Symptom Inventory; MDS = myelodysplasia; PROMS = patient-reported outcome measures; RCT(s) = randomized controlled trial(s); SF-36 = 36-Item Short Form Health Survey.

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Funded by NHS Blood and Transplant via the Systematic Review Initiative.

Received for publication April 3, 2013; revision received July 18, 2013, and accepted July 30, 2013.

doi: 10.1111/trf.12441

TRANSFUSION 2014;54:1166-1179.

supportive care),[2] usually when their PLT count is less than 10×10^9/L unless they have other risk factors for bleeding.[2,3] There have been two recent large randomized controlled trials (RCTs) that compared the use of prophylactic PLT transfusions versus therapeutic-only PLT transfusions.[4,5] Both showed that prophylactic PLT transfusions reduced bleeding (World Health Organization [WHO] Grade 2 bleeding or above 43% vs. 50%),[5] but this strategy required significantly more PLT transfusions (mean number/patient, 3 vs. 1.7; $p < 0.001$).[5] It is essential to determine whether any benefits of PLT transfusions (e.g., reduced bleeding and patient distress) are outweighed by their costs (e.g., risks of transfusion). One way to measure patient distress would be via measurement of health-related quality of life (HRQoL) or illness perceptions or representations (IPs).

Recent data from the EXTEND[6] and RAISE[7] RCTs has shown that bleeding has a significant impact on HRQoL in patients with immune thrombocytopenia (ITP).[8] However, no PLT transfusion study within two recent systematic reviews assessed patients' HRQoL or IPs as an outcome measure.[9,10] The majority of bleeding seen within these PLT transfusion studies is considered minor or moderate by researchers and clinicians (WHO Grade 1 or 2).[4,11,12] Also, skin bleeding is often not considered clinically significant.[4,12-15] These trials used bleeding scales to produce a more objective assessment of bleeding; however, these scales do not take into account the impact bleeding may have on the patients' "quality of life." For example, a new bleeding grading system (Bleeding Severity Measurement Scale) did not consider a nosebleed that did not require nasal packing or a transfusion as clinically significant.[12] However, for patients with hematologic malignancies or MDS we do not know what effect this minor or moderate bleeding has on patients' symptom distress or HRQoL. Patients with ITP differ significantly from patients with hematologic malignancies or MDS. The main symptom patients with ITP will have is bleeding, whereas patients with hematologic malignancies or MDS often have a plethora of other symptoms due to their diagnosis, treatment, or coexisting illnesses. The incremental impact of bleeding (usually WHO Grade 1 or 2) on HRQoL or symptom distress in patients with hematologic malignancies or MDS may be minimal.

HRQoL refers to how an individual's well-being (all emotional, social, and physical aspects of the individual's life) may be impacted over time by a disease, a disability, or a disorder.[16] It is inherently subjective and is best measured from the patient's perspective.[1] Because it is multidimensional, its measurement requires the investigator to ask about a range of areas of the patient's life, including physical, emotional, and social well-being as well as functional ability.[1] HRQoL is now usually measured quantitatively using questionnaires (patient-reported outcome measures [PROMS]).[16] These questionnaires should meet certain quality criteria regarding their reliability and validity in the patient population being studied.[17] They can be divided into two main types: generic instruments (e.g., 36-Item Short Form Health Survey [SF-36]);[18] and disease- or disorder-specific instruments (e.g., Functional Assessment of Cancer Therapy–Bone Marrow Transplantation [FACT-BMT]).[19]

IPs are patients' beliefs and expectations about an illness or physical symptom. They are central to Leventhal's Common-Sense Model (CSM) of Illness Representation (also called Self-Regulation Theory).[20-22] The Common-Sense Model identifies the factors involved in the patient's processing of information regarding their disease or illness, how this information is integrated to provide a "lay" view of the illness, and how this lay view guides coping behaviors and outcomes (Fig. 1).[23] Research has identified six cognitive components of IPs (these will be referred to as perceptions throughout the rest of the review).[21,22,24]

A popular quantitative tool used to elicit IPs is the Illness Perception Questionnaire (IPQ).[25] This has been revised to include emotional representations and illness coherence (Revised Illness Perception Questionnaire).[26] Across the questionnaires, high scores on Consequences, Identity, Emotional Representations, Timeline Acute or Chronic, and Timeline Cyclical dimensions typically represent more negative beliefs. Conversely, high scores on the Coherence, Personal Control, and Treatment Control dimensions represent more positive beliefs.[27]

Why it is important to perform this review

There is a need to understand the effect, if any, of bleeding on patients' IPs and HRQoL and ultimately upon clinical outcomes. This would enable a more robust evaluation of PLT transfusion strategies and disease treatments in the future.

Objectives

The primary objective was to assess whether bleeding affects HRQoL or IPs in patients with hematologic malignancies or MDS. The secondary objectives were to determine methodologic quality of included studies and explore any difference in patient-reported outcomes with regards to bleeding, whether quantitative measures of HRQoL or IP have been developed specifically related to bleeding, and whether any studies have used interventions to reduce symptom distress.

MATERIALS AND METHODS

Protocol and registration

Full details of the inclusion criteria and methods for the analysis were pre-specified and documented in a protocol

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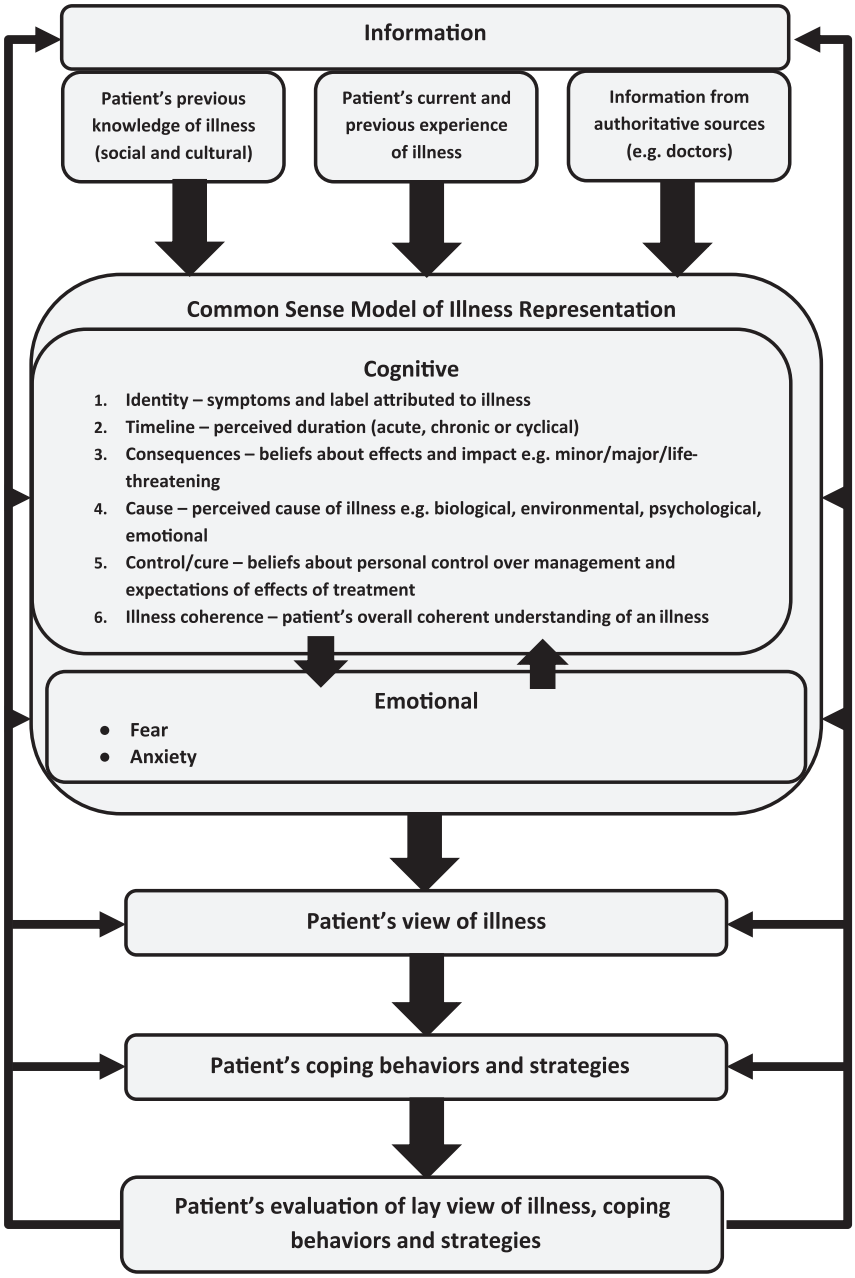

Fig. 1. Leventhal's common sense model of illness representation.

(PROSPERO [CRD 42011001446] and published online [http://www.crd.york.ac.uk/prospero/]).

Criteria

Types of studies

RCTs, prospective cohort studies, and prospective cross-sectional studies were included in this review. These studies could be quantitative or qualitative in nature. Retrospective studies, case reports, animal studies, laboratory studies, and reviews were excluded from the review.

Types of participants

Patients of any age were included if they had MDS or a hematologic malignancy and had severe thrombocytopenia ($\leq 50 \times 10^9$/L for at least 5 days) or were expected to develop severe thrombocytopenia due to their treatment. HRQoL or IP evaluation and bleeding had to be primary or secondary outcomes in the original study. We excluded studies that only reported bleeding as an adverse event. Any study that only reported bleeding as an adverse event would not separate patients with bleeding and those who did not have bleeding into two separate groups and compare PROMS between the two groups.

Studies were excluded if patients had other hematologic disorders, patients did not have thrombocytopenia, or only family members or caregivers were asked about HRQoL. If studies consisted of mixed populations of patients, with diagnoses of solid tumors or nonmalignant hematologic disorders, only data from the hematologic malignancy or MDS subgroups were included. If subgroup data were not provided (after the study authors were contacted), the study was excluded if fewer than 80% of participants had hematologic malignancies or MDS.

Types of outcome measures

Primary outcomes

- HRQoL or IPs and its relation to bleeding

Secondary outcomes

- Number and severity of bleeding episodes
- Mortality (all cause and secondary to bleeding)

Search methods

Nine electronic databases (including MEDLINE [1948 to 2013]; EMBASE [1980 to 2013]; CINAHL [1982 to 2013];

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Transfusion Evidence Library [1980 to 2013]), as well as three electronic trials databases, were searched in full up to January 7, 2013. There was no restriction on language or publication period (see Appendix S1 for search strategies, available as supporting information in the online version of this paper). Database searching was augmented by hand-searching reference lists of all included studies, relevant review articles, and current treatment guidelines.

Data collection and analysis

Selection of studies

All electronically derived citations and abstracts of papers identified by the review search strategy were initially screened by one reviewer (CD) and duplicates were removed. The abstracts were then screened for relevancy by another reviewer (LE, AK, or DP). Twenty percent of these abstracts were selected randomly and these were then screened independently by a third reviewer (LE, AK, or DP): any disagreements were resolved by discussion. Studies not meeting the inclusion criteria were excluded at this stage. The full text of all potentially relevant studies were formally assessed for eligibility by any two out of four independent reviewers (LE, AK, DP, ES): any disagreements were resolved by discussion. Further information was sought from the study authors when the study reports contained insufficient data to enable a decision to be made about eligibility for this review. All studies that failed to meet our eligibility criteria were detailed in a table of excluded studies. Methodologic quality was not an exclusion criterion.

Data extraction and management

Data extraction was conducted according to the guidelines proposed by The Cochrane Collaboration,[28] by two authors working independently (DP, ES). Disagreements between the review authors were resolved by consensus. A third review author (LE) gave her opinion if the first two authors could not reach a consensus. The review authors were not blinded to names of authors, institutions, journals, or the outcomes of the studies. Data were extracted using a standardized data extraction form designed specifically for use in this review (Appendix S2, available as supporting information in the online version of this paper). When study reports did not provide sufficient information, authors and study groups were contacted for additional details.

Assessment of risk of bias

Any thorough systematic review needs to assess the methodologic quality of the included studies. In systematic reviews, the main issue is whether the effect seen in the primary studies is the true effect or whether a defect in study design (internal validity) has led the result to be over or underestimated (biased).[29]

All included studies were assessed by two review authors (LE, DP) for possible risk of bias. For RCTs this was conducted as described in the Cochrane Handbook of Systematic Reviews of Interventions.[30] The assessment included information about the design, conduct, and analysis of the trial.

For nonrandomized trials no validated critical appraisal or risk of bias tool was available for use.[31-33] Therefore, the risk of bias was assessed using the domains defined by Sanderson and coworkers[31] as critical for the assessment of study quality of observational studies. Sanderson and colleagues developed a tool for assessing whether critical appraisal tools within their systematic review contained the essential information for appraising the conduct of observational studies based on the STROBE guidelines[34] (Appendix S2).

Information from relevant reporting guidelines (CHERRIES, COREQ, STROBE)[34-36] were used to aid the decision on whether an aspect of a particular study design would affect the internal validity of the study. The types of bias were then rated within each category as high, low, or unclear risk of bias. This system of reporting bias is obviously flawed and so the reasons why studies were categorized as low, high, or unclear risk of bias were documented to enable readers to come to their own conclusions about the degree of bias inherent within each study and to aid transparency of the review process.[29]

Measures of treatment effect

When available we recorded dichotomous outcomes as the numbers of outcomes and continuous outcomes as means and standard deviations (SDs). No relative effect measures were calculated. Qualitative information was transcribed from the study report and quoted verbatim.

Dealing with missing data

Three of the four included study authors were contacted by e-mail to obtain information that was missing or unclear in the published reports.[37-39] Two authors responded and provided further unpublished information.[37,38] The author of one of the ongoing studies confirmed that this has not yet been published in full.[40]

Two trials included patients with hematologic malignancies as well as patients with solid tumors or nonmalignant hematologic disorders. The authors were contacted because data could not be extracted for a malignant hematology "subgroup" from the reported information. In both studies, patients with hematologic malignancies constituted only a minority of all study patients and the authors were unable to provide additional information on the hematology subgroup; these studies were therefore excluded from the review.[41,42]

Data synthesis

We provided a narrative synthesis of the findings from the included studies, structured around the type of study

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design and type of outcome. No meta-analyses were performed because the studies were not homogeneous in their study design. No formal assessments of heterogeneity or reporting biases were performed because no meta-analyses were performed.

RESULTS

Description of studies

Results of the search (see Fig. 2—PRISMA flow diagram)

A total of 6247 studies were initially identified after duplicates were removed; 5945 studies were excluded on the basis of the abstract by one review author (LE, AK, or DP) and 20% of these abstracts were randomly selected and independently assessed by a second review author (LE, AK, or DP). A total of 302 full-text articles were assessed for eligibility by any two of four review authors (LE, AK, DP, ES).

Included studies

Study type. There were six studies within seven citations eligible for inclusion within this review. One of these studies is due to start recruiting patients,[43] and the other has not yet been published in full[40] (Table S1 Appendix S5, available as supporting information in the online version

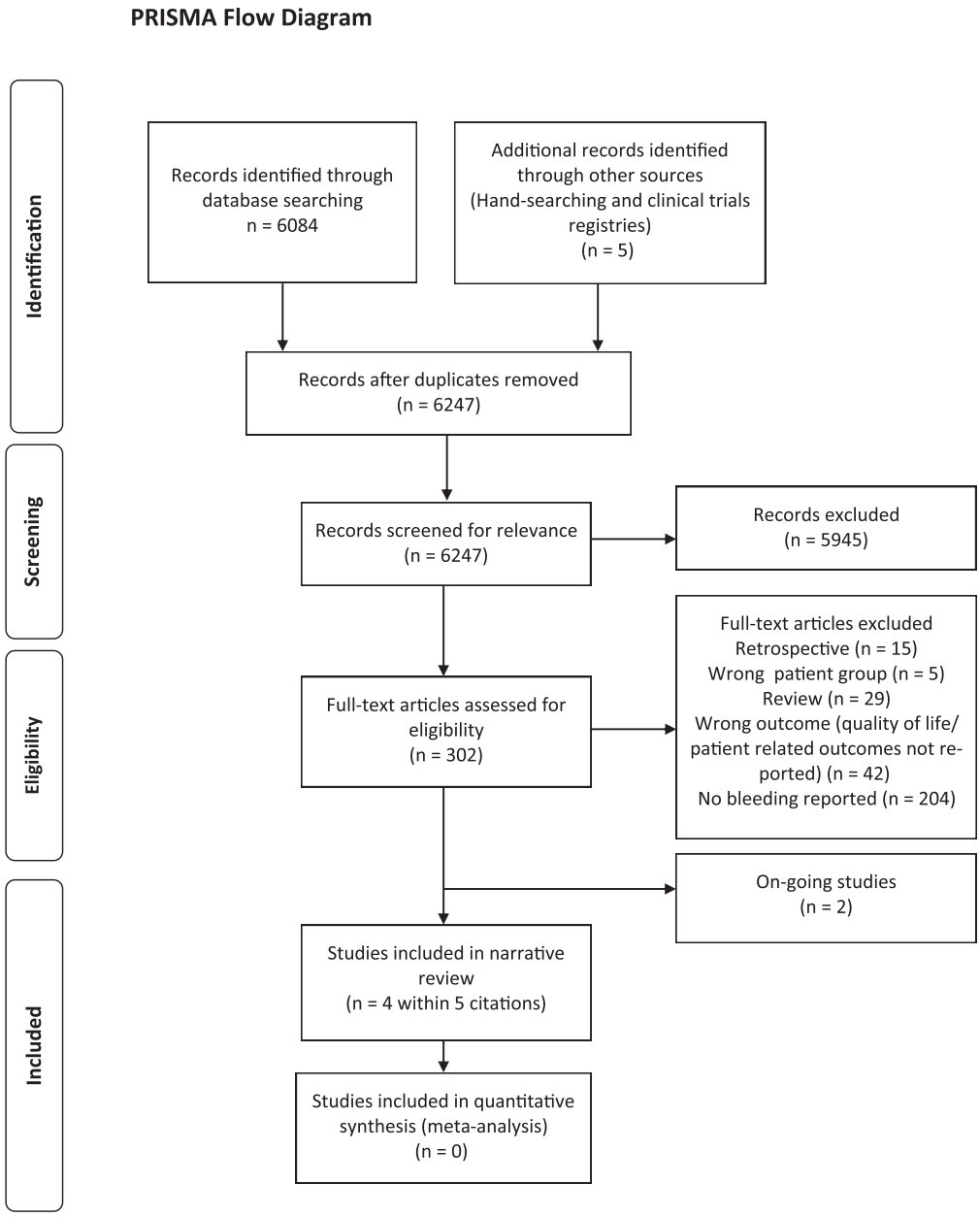

Fig. 2. PRISMA flow diagram.

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of this paper). Of the completed studies, there was one RCT,[39] one prospective observational study,[38] one interview study,[44] and one Web-based survey[37,45] (Table 1). In total four studies were included in this review.

Participants. The four included studies contained 450 participants. One study consisted of only patients with MDS;[37] the other three studies consisted of patients with a mixture of diagnoses (Table 1). The mean age of patients within the studies ranged from 49 to 70 years of age.

Interventions. There was only one interventional study,[39] and this was an RCT. This examined the effects of a computer-assisted, interactive tailored patient assessment (ITPA) tool on patient care, symptom distress, and patients' need for symptom management during treatment and rehabilitation.

Outcomes. None of the four studies were designed to study the effect of bleeding on HRQoL or IPs. All of the studies reported overall symptom distress, disease burden or HRQoL related to their disease, or its treatment as their main outcome measure.[37-39,44] Assessment of symptom distress or disease burden was assessed as a one-off measure in two studies.[37,44] It was assessed over 30 days in one study[38] and for up to 1 year in the final study[39] (Table 1).

Excluded studies

There were 295 studies excluded on the basis of a full text review (Appendix S3, screening algorithm, available as supporting information in the online version of this paper). The reasons for exclusion were as follows: no bleeding reported (n = 204), no patient-related outcome reported (n = 42), review rather than primary study (n = 29), retrospective study (n = 15), and wrong patient group (i.e., not patients with hematologic malignancies or MDS; n = 5; for a list of excluded studies, see Appendix S4, available as supporting information in the online version of this paper).

Risk of bias

All included studies were assessed by two review authors (LE, DP) as previously described and all four studies were found to be at significant risk of bias (see Table 2 for the separate risk categories and reasons why each study was rated as high, low, or unclear risk).

PROMS

IP questionnaires

No IP questionnaires were used to assess patients' IPs within any of the four studies.

HRQoL questionnaires

A wide number of established and novel tools were used to assess the patients' HRQoL within the four included studies (Table 1 and Appendix S5 Table S2). The number of HRQoL assessment tools used in each study varied between one[39] and three.[37,38,44]

The FACT was the only tool used in more than one study.[37,38] Other widely used and validated HRQoL tools within the studies included SF-36,[39] EORTC QLQ-C30,[44] Profile of Mood States,[38] and EuroQol EQ-5D.[37] Two of the studies used novel HRQoL tools that included specific questions relating to bleeding.[37,38] (Table 1 and Appendix S5 Table S2).

Timing of HRQoL assessments (Table 1). All studies completed a minimum of one HRQoL assessment, with the two cross-sectional studies only completing a single assessment.[37,44]

Other PROMs

The RCT[39] used their novel ITPA tool to assess symptom distress and patients' need for symptom management before inpatient and outpatient assessments for up to 365 days.

Assessment of bleeding (see Table 1)

Formalized bleeding assessments

None of the studies reported using a specific bleeding assessment tool.[37-39,44]

Effects of interventions (see Table 3)

Primary outcome: HRQoL or IP and its relation to bleeding

None of the studies reported a quantitative relationship between IP and bleeding or between HRQoL and bleeding.[37-39,44] The RCT only reported a combined outcome measure of bleeding and infection as one of their symptom distress measures.[39] The authors were contacted for further information but did not respond.

In the prospective observational study (autologous stem cell transplant patients) only 3% of patients had moderate or severe bleeding during the study.[38] The authors did not analyze separately patients who bled and patients who did not bleed; it was therefore not possible to assess whether bleeding caused symptom distress (unpublished information from authors).[38]

The online survey of MDS patients assessed concern about bleeding but did not relate it to whether patients had experienced recent bleeding or not.[37] Only 68.5% of patients (137/200) answered this section of the survey, and only 20% of these patients (28/137) had a PLT count of less than 50×10^9/L (Table 4). Significant differences in PLT count groups were obtained on worry about serious bleeding, pinpoint bleeding, and vaginal bleeding; patients within the 30 to 49 PLT count group (7% of patients [9/137]) reported the most problems

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TABLE 1. Characteristics of included studies

Study and country	Study design	Number of participants	Main aim of study	Assessment of patient-related outcomes	Assessment of bleeding	Hematologic diagnosis and treatment of participants	Mean age participants (years)
Ruland 2010[39] Norway	RCT Multicenter	145 (75 intervention, 70 control) 100 followed up for 100 days (19 [13%] died, 10 [7%] dropped out, 16 [11%] lost to follow-up) 51 followed up for 365 days	Does use of a novel ITPA reduce symptom burden	SF-36 (baseline only) ITPA* (before inpatient and outpatient assessments for up to 365 days)	Self-reported using ITPA and retrospective review of the notes to confirm symptoms reported by patient	Leukemia (28; 19%) Lymphoma (111; 77%) Myeloma (6; 4%) Chemotherapy (98; 68%) SCT (47; 32%)	49.5
Anderson 2007[38] United States	Prospective observational study Single center	100 12 lost to follow-up (12%)	Assess symptom burden of patients undergoing autologous SCT	MDASI-BMT* (5× baseline to Day 30) FACT-BMT (baseline and Day 30) POMS (baseline and Day 30)	Method not reported	Non-Hodgkin's lymphoma (34) Myeloma (66) SCT (100)	53.6
Persson 1995[44] Sweden	Open-ended interview Single center	5	Explore patients' experience of physical and psychological problems during active phase of disease and treatment	Open-ended interview EORTC QLQ-C30 SOC LGC (single assessment)	Noted by interviewer. No systematic assessment of bleeding.	Acute leukemia (4; 80%); lymphoma (1; 20%) Active treatment (5; 100%), type of treatment not reported	70.4
List 2009[37] United States	Web-based survey Multicenter	200† 137 included in analysis of bleeding and PLT count†	Explore the patient burden associated with MDS disease and treatment	Bespoke measure* included† EQ-5D FACT-Th FACT-G Other items from questionnaires used in clinical studies MDS (single assessment)	Self-reported	MDS Only 28/137 had PLT count $<50 \times 10^9$/L† Chemotherapy (39/70)‡†	64.7

* PROM devised for this study.
† Unpublished data.
‡ Only 70 of the 200 study patients reported whether or not they were currently receiving any chemotherapy treatment.
EORTC = European Organization for Research and Treatment of Cancer; FACT-G = FACT–General; FACT-Th = FACT–Thrombocytopenia; IPSS = International Prognostic Scoring System; LGC = Lund Gerontology Center; POMS = Profile of Mood States; SCT = stem cell transplant; SOC = Sense of Coherence Scales.

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TABLE 2. Risk of bias table

Study name	Ruland 2010[39]	Anderson 2007[38]	Persson 1995[44]	List 2009[37]
Type of study	RCT	Prospective observational cohort study	Interview study	Online survey
Selection bias	Low risk *Adequate sequence generation* Computer-generated minimization algorithm that equalized groups on gender and type of treatment. *Allocation concealment* Tablet computer was programmed so that the randomization algorithm could only be accessed after completion of baseline assessment.	Unclear risk *Appropriate source population* 100 of 148 eligible patients approached agreed to participate but did not describe how it was decided which patients should be approached.	Unclear risk *Appropriate source population* "A consecutive sample of five patients" with acute leukemia or a high grade lymphoma. Did not describe whether they were consecutive admissions, or in what way they were consecutive.	High risk *Appropriate source population* Selected group of patients. Invited into study on attendance at MDS Foundation forum or via the Internet.
Performance bias	High risk *Blinding of clinician and patient* Due to nature of study the patient's group assignment could not be concealed after baseline data were obtained as it was to provide nurses and physicians with assessment summaries of patient symptoms, problems, and concerns.	Unclear risk *Appropriate methods to deal with any design-specific issues* Self-report by patient; however, research nurses had provided education. Not enough detail provided on education regarding completion of forms.	Unclear risk *Appropriate methods to deal with any design-specific issues* No information on who conducted interview, whether a relationship was established between the interviewer and patient before study commencement, whether the setting for the interview was appropriate, or whether interviewer may be biased.	High risk *Appropriate methods to deal with any design-specific issues* Self-report by patient.
Detection bias	Low risk *Blinding of outcome assessor* To compare groups, two trained raters who were blinded to patients' study group assignment conducted independent chart audits for inpatient and outpatient visits and abstracted both physicians' and nurses' inpatient and outpatient notes for symptoms, problems, and concerns equivalent to those contained in the choice ITPA.	High risk *Blinding of outcome assessor* Only ECOG PS (Eastern Co-operative Oncology Group Performance Status) was used to assess the health care provider's estimate of patient's functional status at each time point. Laboratory variables that might affect patient's symptoms were also recorded. No independent blind assessment of outcome reported.	High risk Patients minimized their physical problems. Weight loss evident in all patients, but only one saw it as a problem. Severe oral complications noted by interviewer, but patients did not talk about them much. No independent report by nurse or family member.	High risk Self-report by patient.
Incomplete outcome data (attrition bias)	High risk Authors mentioned attrition rates, only 51 patients were followed up for 1 year, the planned duration of the study.	High risk Three patients lost to follow-up before blood count nadir. Nine patients lost by the time of last assessment at 30 days (missing data replaced by last observation carried forward).	Unclear risk No discussion of possibility of saturation of data.	High risk 200 patients participated in online study. Only 137 completed section on bleeding and PLT count.*
Selective reporting (reporting bias)	High risk Bleeding and infection assessed under single subgroup of symptom distress.	High risk FACT-BMT p values only reported in study for baseline against symptoms of fatigue, sleep disturbance, lack of appetite, and pain.	Low risk Tape-recorded interview transcribed verbatim and "analyzed from a hermeneutic phenomenological perspective." Independently assessed by two authors. Quotations provided and organized into themes: "physical problems; psychosocial problems; patients' coping strategies . . . ; patients' opinions of the care provided." "Methodological triangulation was applied."	Low risk Unpublished data from MDS Foundation provided.*
Other bias (including funding)	Low risk No other sources of bias identified.	Low risk No other sources of bias identified.	Low risk No other sources of bias identified.	Low risk No other sources of bias identified.
Protocol deviation	Unclear risk No protocol available.	Unclear risk No protocol available.	Unclear risk No protocol available.	Unclear risk No protocol available.

* Unpublished data.
NA = type of bias not applicable to study design.

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TABLE 3. Primary and secondary outcomes of the review

Study	Primary outcome: Quality of life or IP and its relation to bleeding	Secondary outcomes: Number and severity of bleeding episodes	Secondary outcomes: All-cause mortality and mortality secondary to bleeding	Comment
Ruland 2010[39]	NR	NR	19 died during first 100 days: 11 in the intervention group eight in the control group Mortality due to bleeding not reported	Bleeding and infection assessed under single subgroup of symptom distress
Anderson 2007[38]	NR	Percentage of patients with moderate or severe bleeding: 0% baseline 3% count nadir 1% day 30	NR	MDASI-BMT mean symptom score bleeding: 0.14 (SD, 0.62) baseline 0.31 (SD, 1.20) count nadir 0.15 (SD, 0.89) Day 30
Persson 1995[44]	Two patients who had epistaxis during interview did not consider it important	Two patients experienced epistaxis during interview. Interviewer noted bruising. Severity of bleeding was not reported.	NA	
List 2009[37]	NR	Self-reported bleeding in week before survey (Table 4).	NA	Significant differences in PLT count groups were obtained on worry about serious bleeding, pinpoint bleeding and, vaginal bleeding, 30-49 PLT count group (nine patients) reporting most problems.*

* Unpublished data.
NA = not applicable; NR = not reported.

TABLE 4. Bleeding present during preceding week within List 2009[37] (unpublished data)

Bleeding	PLT count ($\times 10^9$/L) <30	30 to 49	50 to 99	>100	Total
Presence or absence of bleeding reported	18/19 (94.7)	7/9 (77.8)	9/23 (39.1)	42/86 (48.8)	137
Spontaneous bleeding	4/18 (22.2)	3/7 (42.9)	0/9 (0)	2/42 (4.8)	76
Bruises	11/18 (61.1)	6/7 (85.7)	3/9 (33.3)	14/42 (33.3)	76
Blood blisters in mouth	1/18 (5.6)	1/7 (14.3)	0/9 (0)	4/42 (9.5)	76
Blood in urine	0/18 (0)	0/7 (0)	0/9 (0)	0/42 (0)	76
Blood in stool	3/18 (16.7)	1/7 (14.3)	2/9 (22.2)	2/42 (4.8)	76

* Data are reported as number/total (%).

(unpublished information provided by the author).[37] The 14% (19/137) of patients with PLT counts below 30 reported far fewer concerns about bleeding. None of the PLT count groups showed differences in concern about bleeding from the mouth and gums or nose or seeing blood in the urine or stool.

In the interview study (leukemia and lymphoma patients), bleeding was only mentioned as part of the open-ended interview, no quantitative assessment was performed.[44] Two of the five patients had nosebleeds (during the interview)[44] and neither participant felt that this was important. However, the interviewers noted that all the patients downplayed all their physical symptoms significantly despite objective evidence of significant weight loss, hair loss, mucositis, or bleeding.

Secondary outcome: number and severity of bleeding episodes

Three of the four studies reported bleeding[37,38,44] (see Table 3). The prospective observational study reported bleeding at PLT count nadir in only 3% of patients.[38] The interview study reported epistaxis in two patients and bruising was noted by the interviewers but there was no comment about the severity of the bleeding.[44] The online survey asked about bleeding present in the preceding week, however, the answers were poorly completed.

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Overall, only 55% of patients (76/137 with a known PLT count) reported whether they had had spontaneous bleeding, bruising, blood blisters in the mouth, or blood in the urine or stool within the previous week (unpublished data provided by the author; Table 4).[37]

Secondary outcome: mortality (all cause or secondary to bleeding)

This outcome was only relevant to two of the four studies[38,39] (longitudinal studies). The other two studies were of a cross-sectional design and all patients had to be alive to participate in the studies.[37,44]

Of these two studies, only one study reported all-cause mortality.[39] Eleven patients died in the intervention arm and eight patients died in the control arm during the first 100 days. Neither study reported mortality due to bleeding.

DISCUSSION

The main objective of this review was to determine whether bleeding affects HRQoL or IP in patients with hematologic malignancies or MDS. Other objectives were to determine whether a HRQoL tool had been developed that specifically assessed patient distress or concern in relation to bleeding symptoms, determine whether any studies had used interventions to reduce symptom distress, and assess the methodologic quality of included studies.

Summary of key findings

Six studies were identified and two of these studies have not been completed. There were therefore four studies included in this review.

This review found no studies that evaluated the differences in HRQoL or IPs between patients who did or did not experience bleeding. Thus, we are unable to establish the extent to which bleeding affects HRQoL or IPs, nor were we able to establish whether patients with MDS had a different perception of bleeding to those patients with transient marrow failure (e.g., after chemotherapy or stem cell transplantation).

The Web-based survey[37] and observational study[38] used two new patient-reported outcome scales FACT-Th (MDS patients) and MD Anderson Symptom Inventory–Bone Marrow Transplant (MDASI-BMT; stem cell transplant patients), respectively, which specifically assessed patient distress or concern in relation to bleeding symptoms. However, these tools were not used within these studies to assess differences in HRQoL due to bleeding. The RCT[39] did use an ITPA tool to assess symptom distress and feedback information to the nurses and doctors but they did not analyze bleeding separately from infection and therefore effectiveness of this tool for reducing symptom distress due to bleeding is unknown.

Quality of the evidence

Overall the quality of the evidence is poor. The four studies identified in the systematic review were all at a significant risk of bias.

There is no universal validated and well-used PROM that could be used to assess the impact of bleeding on quality of life in patients with hematologic malignancies and MDS. The two HRQoL tools that included an assessment of bleeding (MDASI-BMT and FACT-Th) have not been validated in this patient group. Furthermore the remaining HRQoL tools used in these studies were not designed to assess the impact of bleeding upon HRQoL (Table S2 Appendix S5).

In addition the two studies that assessed PROMS to evaluate HRQoL prospectively over time were undertaken over very different time periods. Both studies reported high attrition rates and losses to follow-up.[38,39] The remaining two studies only performed a single assessment (Table 1). HRQoL assessments are more likely to provide meaningful information to health care professionals when they are reported sequentially to provide a representation of changes over time.[46,47]

Patients with hematologic malignancies and MDS often have many other symptoms including side effects of complex and variable treatments, in addition to the psychological impact of fear, insecurity, and caregiver or hospital dependence, all of which are important to patients. Indeed, within the four identified studies patients often rated symptoms other than bleeding as more problematic. However, whether this is due to the underlying study design, in that investigators failed to acknowledge the potential impact of bleeding through direct patient questioning, or whether other symptoms are more concerning to patients is unclear. Certainly, the incidence of bleeding (3%) was much lower in the prospective observational study[38] than rates seen in recent well-conducted PLT transfusion trials of similar patients (45%-59%).[5,48] In these PLT transfusion studies formalized daily bleeding assessments were performed and the rates of severe or life-threatening bleeding were similar to the overall incidence of bleeding (2 to 9%) reported in the cohort study.[38]

Strengths and limitations

We know of no previous reviews on this subject and within this review found four low-quality studies that were not specifically designed to answer the question posed by this systematic review. There were no obvious biases within the review process: a protocol was specified, a wide search was conducted, the relevance of each paper identified was carefully assessed, and no restrictions were made for the language in which the paper was originally published.

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Limiting this review to only patients with hematologic malignancies or MDS may have limited the generalizability of this review to patients with other types of cancer. However, patients with hematologic disorders have more profound and prolonged thrombocytopenia than patients with other types of cancer. This is reflected by the fact that hematology patients use up to 60% of all PLT transfusions issued.[49] We limited our review to patients with hematologic malignancies or MDS because if there is a true difference in symptom distress or HRQoL due to bleeding it is more likely to be detected within this high-risk patient group.

It is important to distinguish between quality of reporting and quality of what was actually done in the design, conduct, and analysis of a study. A high-quality report ensures that all relevant information is available to the reader but does not guarantee it is free of bias.[50]

The assessment of methodologic quality and risk of bias was limited by the lack of a "gold standard" critical appraisal tool. Several systematic reviews have assessed tools for assessing quality and susceptibility to bias in nonrandomized studies.[31-33] All these reviews have concluded that there is no currently agreed gold standard critical appraisal tool; the majority of tools did not undergo a rigorous development process and there are many tools to choose from. Katrak and colleagues[32] identified 121 critical appraisal tools and Sanderson and colleagues[31] identified 86 tools. Quality scales and resulting scores tend to combine aspects of reporting quality and aspects of trial quality, which can lead to a flawed assessment, and therefore these types of assessments should be avoided.[51,52]

Due to these problems, we tried to make our risk of bias assessment as transparent as possible. We used categories that had been previously recognized as critically important for assessing methodologic quality,[31] and provided the reasons we classified a study as at high, unclear, or low risk of bias as well as our summary decision for each domain.

Implications for practice and research

There is a lack of evidence to establish if bleeding is a significant clinical problem that affects patients' HRQoL or IPs. The four studies identified in the systematic review did not focus primarily on bleeding. This review will therefore not affect current clinical practice because we found no evidence on which to base a change in practice.

The main implications of this review are the research agenda, as well as awaiting the results of the two ongoing studies in patients with MDS (Table S1 Appendix S5). Studies are required to assess whether bleeding causes patients distress and, if so, to what extent this occurs and whether this differs between patients with prolonged periods of thrombocytopenia, for example, MDS or patients with reversible thrombocytopenia (e.g., after chemotherapy or a stem cell transplant). Different research approaches may be required, including the development of a validated patient-reported outcome measure, as well as independently verified documentation of the presence or absence of bleeding.

ACKNOWLEDGMENTS

We thank Mr JA Bunola, Consultant Orthopaedic Surgeon, Hull and East Yorkshire NHS Trust; Dr Martin Besser, Consultant Haematologist, Cambridge University Hospitals NHS Foundation Trust; Mr Jean-Christophe Bodhuin; and Mrs Ruth Strachan for their assistance in translating foreign language articles. We thank the authors who provided further unpublished information for their kind cooperation: Karen Anderson, M.D. Anderson Cancer Center; Benjamin Arnold, FACIT organization; James Twiss and Galen Research; and the MDS Foundation. Author contributions: LE—protocol development, searching, selection of studies, eligibility and quality assessment, data analysis, and content expert; DP—protocol development, searching, selection of studies, eligibility and quality assessment, data extraction and analysis, and content expert; ES—selection of studies, eligibility and quality assessment, data extraction, and analysis; AK—protocol development, selection of studies, and eligibility assessment; SB—protocol development and methodologic expert; CD—protocol development and searching and selection of studies; LG—protocol development and content expert; SS—protocol development and content expert; LE drafted the paper and all review authors contributed to the preparation of the final review article.

CONFLICT OF INTEREST

The authors report no conflicts of interest or funding sources.

REFERENCES

1. Cella DF. Quality of life: concepts and definition. J Pain Symptom Manage 1994;9:186-92.
2. Estcourt LJ, Birchall J, Lowe D, et al. Platelet transfusions in haematology patients: are we using them appropriately? Vox Sang 2012;103:284-93.
3. BCSH. British Committee for Standards in Haematology. Guidelines for the use of platelet transfusions. Br J Haematol 2003;122:10-23.
4. Wandt H, Schaefer-Eckart K, Wendelin K, et al. Therapeutic platelet transfusion versus routine prophylactic transfusion in patients with haematological malignancies: an open-label, multicentre, randomised study. Lancet 2012;380: 1309-16.
5. Stanworth SJ, Estcourt LJ, Powter G, et al. The effect of a no-prophylactic versus prophylactic platelet transfusion strategy on bleeding in patients with hematological malignancies and severe thrombocytopenia (TOPPS trial). A randomized controlled, non-inferiority trial. Blood 2012; 120:Abstract 1.

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6. Saleh MN, Bussel JB, Cheng G, et al. Safety and efficacy of eltrombopag for treatment of chronic immune thrombocytopenia: results of the long-term, open-label EXTEND study. Blood 2013;121:537-45.
7. Cheng G, Saleh MN, Marcher C, et al. Eltrombopag for management of chronic immune thrombocytopenia (RAISE): a 6-month, randomised, phase 3 study. Lancet 2011;377:393-402.
8. Fogarty PF, Tarantino MD, Brainsky A, et al. Selective validation of the WHO Bleeding Scale in patients with chronic immune thrombocytopenia. Curr Med Res Opin 2012;28: 79-87.
9. Butler C, Doree C, Estcourt L, et al. Pathogen-reduced platelets for the prevention of bleeding (protocol). Cochrane Database Syst Rev 2013;(3):CD009072.
10. Estcourt L, Stanworth S, Doree C, et al. Prophylactic platelet transfusion for prevention of bleeding in patients with haematological disorders after chemotherapy and stem cell transplantation. Cochrane Database Syst Rev 2012;(5): CD004269.
11. Heddle NM, Arnold DM, Webert KE. Time to rethink clinically important outcomes in platelet transfusion trials. Transfusion 2011;51:430-4.
12. Webert KE, Arnold DM, Lui Y, et al. A new tool to assess bleeding severity in patients with chemotherapy-induced thrombocytopenia. Transfusion 2012;52:2466-74.
13. Rebulla P, Finazzi G, Marangoni F, et al. The threshold for prophylactic platelet transfusions in adults with acute myeloid leukemia. Gruppo Italiano Malattie Ematologiche Maligne dell'Adulto. N Engl J Med 1997; 337:1870-5.
14. Tinmouth A, Tannock IF, Crump M, et al. Low-dose prophylactic platelet transfusions in recipients of an autologous peripheral blood progenitor cell transplant and patients with acute leukemia: a randomized controlled trial with a sequential Bayesian design. Transfusion 2004; 44(12):1711-9.
15. Estcourt LJ, Heddle N, Kaufman R, et al. The challenges of measuring bleeding outcomes in clinical trials of platelet transfusions. Transfusion 2013;53:1531-43.
16. Centers for Disease Control and Prevention (CDC). Health-Related Quality of Life. [cited 2012 Nov 8]. Available from: http://www.cdc.gov/hrqol/
17. Farnik M, Pierzchala WA. Instrument development and evaluation for patient-related outcomes assessments. Patient Relat Outcome Meas 2012;3:1-7.
18. Rand Health. Medical outcomes study: 36-item short form survey. [cited 2012 Nov 8]. Available from: http://www.rand.org/health/surveys_tools/mos/mos_core_36item.html
19. Functional Assessment of Chronic Illness Therapy (FACIT). Functional Assessment of Cancer Therapy—Bone Marrow Transplantation (FACT—BMT). 2007 [cited 2012 Nov 8]. Available from: http://www.facit.org/FACITOrg/Questionnaires
20. Leventhal H. Findings and theory in the study of fear communications. Adv Exp Soc Psychol 1970;5: 119-86.
21. Leventhal H, Meyer D, Nerenz DR. The common sense representation of illness danger. In: Rachman S, editor. Medical psychology. Vol. II. Oxford, UK: Pergamon Press; 1980. p. 7-30.
22. Leventhal H, Diefenbach MA, Leventhal EA. Illness cognition: using common sense to understand treatment adeherence and affect cognition interactions. Cognit Ther Res 1992;16:143-63.
23. Hagger MS, Orbell S. A meta-analytic review of the common-sense model of illness representations. Psychol Health 2003;18:141-84.
24. Lau RR, Hartman KA. Common sense representations of common illnesses. Health Psychol 1983;5: 119-86.
25. Weinman J, Petrie KJ, Moss-Morris R, et al. The Illness Perception Questionnaire: a new method for assessing the cognitive representation of illness. Psychol Health 1996;11: 431-45.
26. Moss-Morris R, Weinman J, Petrie KJ, et al. The revised illness perception questionnaire (IPQ-R). Psychol Health 2002;17:1-16.
27. Mc Sharry J, Moss-Morris R, Kendrick T. Illness perceptions and glycaemic control in diabetes: a systematic review with meta-analysis. Diabet Med 2011;28: 1300-10.
28. Higgins JP, Deeks JJ. Selecting studies and collecting data. In: Higgins JP, Green S, editors. Cochrane handbook for systematic reviews of interventions. Version 5.1.0. The Cochrane Collaboration, 2011. [cited 2013 Sep 13]. Available from: www.cochrane-handbook.org
29. Higgins JP, Altman DG, Gøtzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011;343:d5928.
30. Sterne J, Egger M, Moher D. Addressing reporting biases. In: Higgins JP, Green S, editors. Cochrane handbook for systematic reviews of interventions. Version 5.1.0. The Cochrane Collaboration, 2011. [cited 2013 Sep 13]. Available from: www.cochrane-handbook.org
31. Sanderson S, Tatt ID, Higgins JP. Tools for assessing quality and susceptibility to bias in observational studies in epidemiology: a systematic review and annotated bibliography. Int J Epidemiol 2007;36:666-76.
32. Katrak P, Bialocerkowski AE, Massy-Westropp N, et al. A systematic review of the content of critical appraisal tools. BMC Med Res Methodol 2004;4:22.
33. Crowe M, Sheppard L. A review of critical appraisal tools show they lack rigor: alternative tool structure is proposed. J Clin Epidemiol 2011;64:79-89.
34. von Elm E, Altman DG, Egger M, et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. PLoS Med 2007;4:e296.

15372995, 2014, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.12441 by Deakin University, Wiley Online Library on [25/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

35. Eysenbech G. Improving the quality of web surveys: the checklist for reporting results of internet e-surveys (CHERRIES). J Med Internet Res 2004;6:e34.
36. Tong A, Sainsbury P, Craig J. Consolidated criteria for reporting qualitative research (COREQ): a 32-item checklist for interviews and focus groups. Int J Qual Health Care 2007;19:349-57.
37. List AF, Mathias S, Crosby RD, et al. Disease burden and treatment impact associated with myelodysplastic syndromes: preliminary findings from a web-based survey. ASH Annu Meet Abstr 2009;114:4522.
38. Anderson KO, Giralt SA, Mendoza TR, et al. Symptom burden in patients undergoing autologous stem-cell transplantation. Bone Marrow Transplant 2007;39: 759-66.
39. Ruland C, Holte H, Roislien J, et al. Effects of a computer-supported interactive tailored patient assessment tool on patient care, symptom distress, and patients' need for symptom management support: a randomized clinical trial. J Am Med Inform Assoc 2010;17:403-10.
40. McKenna SP, Twiss J, Wilburn JN, et al. Investigation of the impact of myelodysplasia (MDS) from the patients' perspective. ASH Annu Meet Abstr 2011;118:5049.
41. Feyer P, Kleeberg UR, Steingraber M, et al. Frequency of side effects in outpatient cancer care and their influence on patient satisfaction—a prospective survey using the PASQOC questionnaire. Support Care Cancer 2008;16: 567-75.
42. Arnold B, Dhar J, Parks-Vernizzi E, et al. Translation and linguistic validation of the FACT-Th18 for use with cancer patients with thrombocytopenia worldwide. Value Health 2012;15:A236.
43. ClinicalTrials.gov. NCT01615146. Outpatient Platelet Transfusions in Myelodysplastic Syndromes and Leukemia: the OPTIMAL Pilot. 2012. [cited 2013 Jan 7]. Available from: http://clinicaltrials.gov/show/NCT01615146
44. Persson L, Hallberg IR, Ohlsson O. Acute leukaemia and malignant lymphoma patients' experiences of disease, treatment and nursing care during the active treatment phase: an explorative study. Eur J Cancer Care (Engl) 1995; 4:133-42.
45. On behalf of the MDS Foundation and its Board of Directors. P097 Disease burden and treatment impact associated with myelodysplastic syndromes: initial estimates. Leuk Res 2009;33(Suppl 1):S115-6.
46. Crosby RD, Kolotkin RL, Williams GR. Defining clinically meaningful change in health-related quality of life. J Clin Epidemiol 2003;56:395-407.
47. Juniper EF, Guyatt GH, Willan A, et al. Determining a minimal important change in a disease-specific Quality of Life Questionnaire. J Clin Epidemiol 1994;47: 81-7.
48. Slichter SJ, Kaufman RM, Assmann SF, et al. Dose of prophylactic platelet transfusions and prevention of hemorrhage. N Engl J Med 2010;362:600-13.
49. Pendry K, Davies T. An audit of the use and wastage in the North West of England and North Wales—where have all the platelets gone? Blood Transfus Matters 2011;34:17-9.
50. Huwiler-Muntener K, Juni P, Junker C, et al. Quality of reporting of randomized trials as a measure of methodologic quality. J Am Med Assoc 2002;287:2801-4.
51. Juni P, Witschi A, Bloch R, et al. The hazards of scoring the quality of clinical trials for meta-analysis. J Am Med Assoc 1999;282:1054-60.
52. Greenland S, O'Rourke K. On the bias produced by quality scores in meta-analysis, and a hierarchical view of proposed solutions. Biostatistics 2001;2:463-71.
53. The EuroQoL Group. EuroQol—a new facility for the measurement of health-related quality of life. The EuroQol Group. Health Policy 1990;16:199-208.
54. Brooks R. EuroQol: the current state of play. Health Policy 1996;37:53-72.
55. Jakobsson U, Hallberg IR, Westergren A. Exploring determinants for quality of life among older people in pain and in need of help for daily living. J Clin Nurs 2007;16(3A): 95-104.
56. Lorr M, McNair DM, Heuchert JW, et al. POMS Profile of Mood States. 2004.
57. Bowling A. Measuring disease. 2nd ed. Oxford, UK: Oxford University Press; 2001.
58. Nyenhuis DL, Yamamoto C, Luchetta T, et al. Adult and geriatric normative data and validation of the profile of mood states. J Clin Psychol 1999;55:79-86.
59. Hays RD, Sherbourne CD, Mazel RM. User's manual for the medical outcomes study (MOS) core: measures of health-related quality of life. Santa Monica (CA): RAND Corporation; 1994.
60. Eriksson M, Lindstrom B. Validity of Antonovsky's sense of coherence scale: a systematic review. J Epidemiol Community Health 2005;59:460-6.
61. Bowling A. Measuring health. 2nd ed. Oxford, UK: Oxford University Press; 1997.
62. Persson L, Hallberg IR, Ohlsson O. Survivors of acute leukaemia and highly malignant lymphoma—retrospective views of daily life problems during treatment and when in remission. J Adv Nurs 1997;25:68-78.
63. Aaronson NK, Ahmedzai S, Bergman B, et al. The European Organization for Research and Treatment of Cancer QLQ-C30: a quality-of-life instrument for use in international clinical trials in oncology. J Natl Cancer Inst 1993;85: 365-76.
64. Hjermstad MJ, Fayers PM, Bjordal K, et al. Health-related quality of life in the general Norwegian population assessed by the European Organization for Research and Treatment of Cancer Core Quality-of-Life Questionnaire: the QLQ = C30 (+3). J Clin Oncol 1998;16:1188-96.
65. Osoba D, Aaronson N, Zee B, et al. Modification of the EORTC QLQ-C30 (version 2.0) based on content validity and reliability testing in large samples of patients with cancer. The Study Group on Quality of Life of the EORTC

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and the Symptom Control and Quality of Life Committees of the NCI of Canada Clinical Trials Group. Qual Life Res 1997;6:103-8.
66. Cella DF, Tulsky DS, Gray G, et al. The Functional Assessment of Cancer Therapy scale: development and validation of the general measure. J Clin Oncol 1993;11:570-9.
67. Webster K, Cella D, Yost K. The Functional Assessment of Chronic Illness Therapy (FACIT) Measurement System: properties, applications, and interpretation. Health Qual Life Outcomes 2003;1:79.
68. Cella D, Beaumont J, Webster K, et al. Measuring the concerns of cancer patients with low platelet counts: the Functional Assessment of Cancer Therapy–Thrombocytopenia (FACT-Th) questionnaire. Support Care Cancer 2006;14:1220-31.
69. Cleeland CS, Mendoza TR, Wang XS, et al. Assessing symptom distress in cancer patients: the M.D. Anderson Symptom Inventory. Cancer 2000;89:1634-46.
70. Cleeland CS. The M.D. Anderson Symptom Inventory User Guide [Draft]. 2010. [cited 2013 Sep 9]. Available from: http://www.mdanderson.org/education-and-research/departments-programs-and-labs/departments-and-divisions/sympton-research/sympton-assessment-tools/MDASI_userguide.pdf

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Appendix S1. Search strategy.
Appendix S2. Data recorded on data extraction form.
Appendix S3. Screening of full text for systematic review QoL/Illness perceptions.
Appendix S4. Excluded studies.
Appendix S5. Additional tables.

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